

数学物理方程



2 分离变量法

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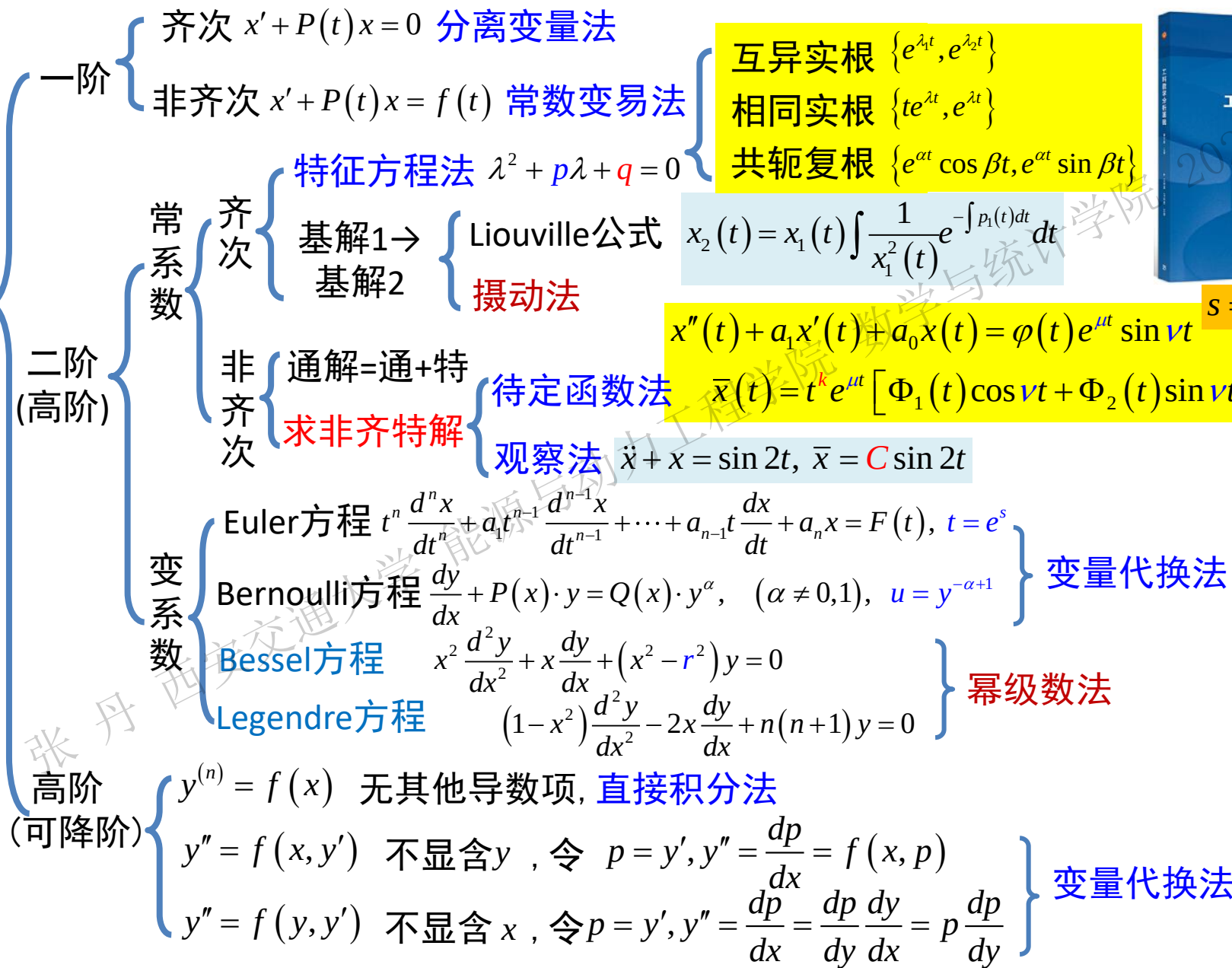
数学与统计学院(兼)

动力工程多相流国家重点实验室

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$

线性常微分方程解法



$$s = \mu \pm \nu i$$

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$



例1 求下列齐次ODE的通解

(1) $y'' - 3y' + 2y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 - 3\lambda + 2 = 0$

特征根 $\lambda_1 = 1, \lambda_2 = 2$

互异实根, 基解组 $\{e^x, e^{2x}\}$

通解 $y = C_1 e^x + C_2 e^{2x}$

(2) $y'' + 4y' + 13y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 13 = 0$

特征根 $\lambda_{1,2} = -2 \pm 3i$

共轭复根, 基解组 $\{e^{-2x} \cos 3x, e^{-2x} \sin 3x\}$

通解 $y = C_1 e^{-2x} \cos 3x + C_2 e^{-2x} \sin 3x$

二阶常系数ODE $x''(t) + px'(t) + qx(t) = 0$

特征方程

$$\lambda^2 + p\lambda + q = 0$$

互异实根 $\{e^{\lambda_1 t}, e^{\lambda_2 t}\}$

相同实根 $\{te^{\lambda t}, e^{\lambda t}\}$

共轭复根 $\{e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t\}$

(3) $y'' + 4y' + 4y = 0$

解 二阶线性常系数ODE

特征方程 $\lambda^2 + 4\lambda + 4 = 0$

特征根 $\lambda_{1,2} = -2$

相同实根, 基解组 $\{e^{-2x}, xe^{-2x}\}$

通解 $y = C_1 e^{-2x} + C_2 x e^{-2x}$

二阶齐次ODE求解的核心是寻找2个基解

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \cdots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$

例2 求下列ODE的通解

(1) $y' - 2y = 0$

解法1 一阶线性常系数ODE

观察法 易知 $y=0$ 是原方程的解

当 $y \neq 0$ 时, 分离变量法

$$\frac{dy}{dx} = 2y \rightarrow \frac{dy}{y} = 2dx \rightarrow \ln y = 2x + C$$

原ODE通解 $y = C \exp(2x)$

$C=0$ 时
兼容 $y=0$

解法2 特征方程法

原ODE对应特征方程 $s - 2 = 0$

特征根 $s = 2$, 对应基解 $\exp(2x)$

即得原ODE通解 $y = C \exp(2x)$

1. 验证0/常数是否为解? 能否被通解包含?

2. 但数学物理方程更加关心的是非0解

(2) $y' - 2y = x + 2$

解 一阶线性非齐次ODE, 常数变易法

变易对应齐次方程通解系数

对应齐次方程的通解左例已求出 $y = C \exp(2x)$

设非齐次方程的通解为 $y = C(x) \exp(2x)$

则 $y' = C'(x) \exp(2x) + 2C(x) \exp(2x)$ 代入原方程

$$C'(x) \exp(2x) + C(x) \cdot 2 \exp(2x) - 2 \cdot C(x) \exp(2x) = x + 2$$

化简得 $C'(x) = (x + 2) \exp(-2x)$ 积分有

$$C(x) = \int (x + 2) \exp(-2x) dx = -\frac{1}{2} \left(x + \frac{5}{2} \right) \exp(-2x) + C$$

原ODE通解为

$$\begin{aligned} y &= C(x) \exp(2x) = \left[-\frac{1}{2} \left(x + \frac{5}{2} \right) e^{-2x} + C \right] \exp(2x) \\ &= C \exp(2x) - \frac{x}{2} - \frac{5}{4} \end{aligned}$$

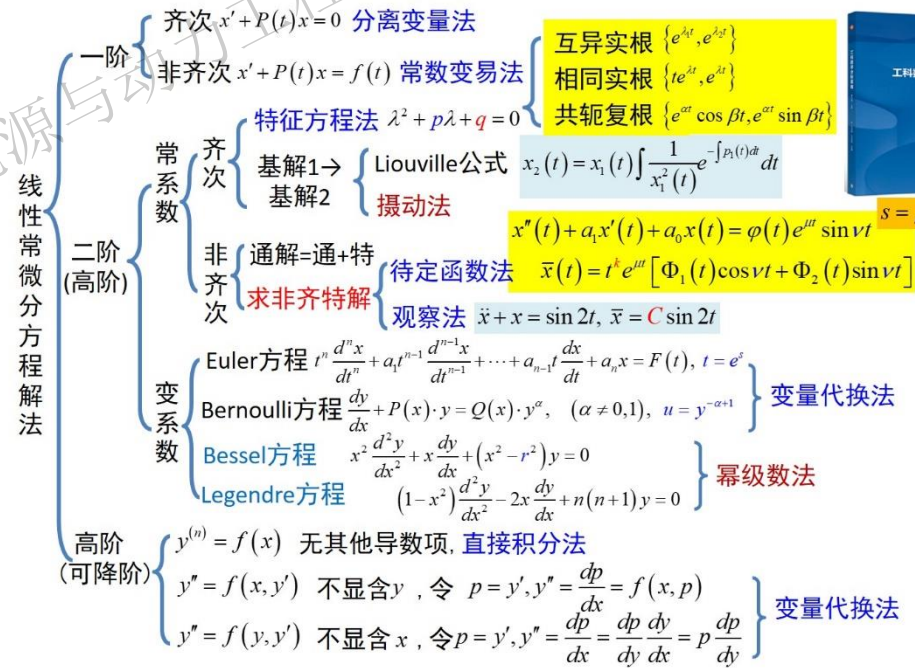
快速计算手册

边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

线性ODE的解法

$$x^{<n>} + P_1 x^{<n-1>} + P_2 x^{<n-2>} + \dots + P_{n-1} x^{<1>} + P_n x^{<0>} = F$$



1. 数学建模和基本原理介绍

2. 分离变量法

2.1 特征值问题

矩阵的特征值

一个二阶微分算子的特征值

2.2 分离变量法

2.3 基于MATLAB的定解 问题求解方法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

为什么要有特征值

1. 数学本质: 物理/社会现象变量关系→抽象提炼定义运算→研究运算→简化运算
2. 运算本质=定义在给定{集合|元素}上的操作
3. 运算多样/对象复杂/结果繁复→如何简化运算?→前提是如何衡量运算/操作效果?

$$\text{操作}(\text{元素}_i) = \lambda \cdot \text{元素}_i \quad \{\text{集合}|\text{元素}_i \text{属于操作对象集合}\}$$

反向评价法 若定义的 操作 对某非零元素无可奈何→即上式存在非零解
→反过来该元素(非零解)可作为特征→衡量操作的作用范围/效果→进而
探析简化运算的途径

$$Ax = \lambda x, \quad x \in R^n$$

特征向量 特征值

2.1 特征值问题

利用熟悉的矩阵特征值/特征向量→正交分解→简化运算

矩阵特征值

n 阶实矩阵 A 左乘列向量是针对 n 维列向量的线性变换

矩阵 A 的特征值是如下方程的非零解

$$Ax = \lambda x, \quad x \in \mathbb{R}^n$$

特征(列)向量 特征值

条件

- 表述1 矩阵 A 有 n 个线性无关的特征向量
- 表述2 特征值的代数重数=特征向量的几何重数

$$T_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{bmatrix}$$

n 个特征向量组成的方阵

$$A[T_1, T_2, \dots, T_n] = [T_1, T_2, \dots, T_n]$$

对角阵右乘方阵=对角元素乘以对应列

$$\begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} = [\lambda_1 T_1, \lambda_2 T_2, \dots, \lambda_n T_n]$$



$$AT_i = \lambda_i T_i$$

$$T = [T_1, T_2, \dots, T_n]$$

左乘运算对特征向量无可奈何

- 正交性←特征向量线性无关
- 完备性 $T = [T_1, T_2, \dots, T_n]$ 可作为 $\mathbb{R}^n$ 的一组基, $\mathbb{R}^n$ 中任意向量均可由该组基表示

对向量的左乘(线性变换)→特征值/特征向量→特征方向上的标量计算(系数计算)

2.1 特征值问题

特征值如何简化运算-方阵左乘

例1 矩阵左乘列向量的计算. 二维矩阵左乘(向量)=旋转+放缩

$$A\mathbf{x} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

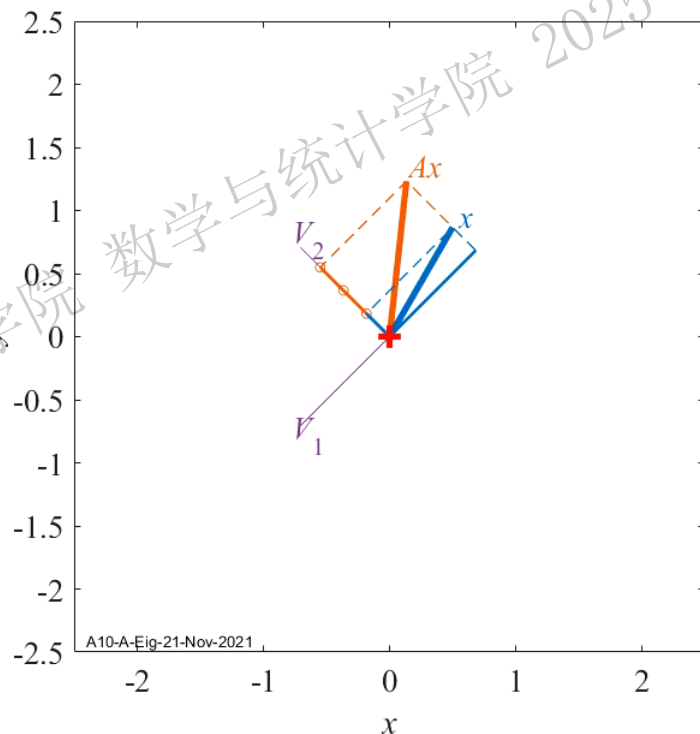
直接法: 按左乘定义计算

间接法: 按特征向量分解

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \in \mathbb{R}^n$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

特征向量 特征值



将 $\mathbf{x}$ 按特征方向 $\mathbf{v}_1, \mathbf{v}_2$ 分解 $\rightarrow$ 每个方向按特征值放缩 $\rightarrow$ 合成得到左乘结果 $A\mathbf{x}$

1. 运算对象 $\rightarrow$ 用特征向量(函数)的线性组合表示 $\rightarrow$ 对象规范化的通用手段
2. 按特征方向分解 $\rightarrow$ 方阵左乘运算 $\rightarrow$ 各特征方向的系数计算 $\rightarrow$ 运算大大简化

2.1 特征值问题

例2 若方阵 A 可对角化, 利用正交分解求解线性方程组 $Ax = b$, $b \in R^n$

解 将 A 的 n 个线性无关的特征向量作为一组基 $\{T_1, T_2, \dots, T_n\}$

将 x, b 按照此基展开 $x = \sum_{i=1}^n x_i T_i$, $b = \sum_{i=1}^n b_i T_i$

正交/完备

原方程可化为 $Ax = b$ 求得即得解

$$A \sum_{i=1}^n x_i T_i = \sum_{i=1}^n b_i T_i$$

$$AT_i = \lambda_i T_i$$

$$\sum_{i=1}^n x_i AT_i = \sum_{i=1}^n b_i T_i$$

$$\sum_{i=1}^n x_i \lambda_i T_i = \sum_{i=1}^n b_i T_i$$

$$x_i \lambda_i = b_i$$

基线性无关, 比较系数可得 $x_i = \frac{b_i}{\lambda_i}$

原方程的解可表示为 $x = \sum_{i=1}^n x_i T_i = \sum_{i=1}^n \frac{b_i}{\lambda_i} T_i$

2.1 特征值问题

例3 若矩阵 A 可对角化, 利用正交分解求解非齐次ODE方程组

$$\frac{dx(t)}{dt} = Ax + f(t), \quad x(t), \quad x(t) = x^0 \in \mathbb{R}^n$$

正交/完备

解 将A 的 n 个线性无关的特征向量作为一组基 $\{T_1, T_2, \dots, T_n\}$

将 x, x^0, f 分别按照此基展开 $x(t) = \sum_{i=1}^n x_i(t) T_i, \quad x^0 = \sum_{i=1}^n x_i^0 T_i, \quad f(t) = \sum_{i=1}^n f_i(t) T_i$

原方程可化为 $\frac{dx(t)}{dt} = Ax + f(t)$ 求得即得解

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = A \sum_{i=1}^n x_i(t) T_i + \sum_{i=1}^n f_i(t) T_i$$

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n x_i(t) AT_i + \sum_{i=1}^n f_i(t) T_i$$

$AT_i = \lambda_i T_i$

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n x_i(t) \cdot \lambda_i T_i + \sum_{i=1}^n f_i(t) T_i$$

每个特征方向上的 ODE 具有相同形式

$$\sum_{i=1}^n \frac{dx_i(t)}{dt} T_i = \sum_{i=1}^n [x_i(t) \cdot \lambda_i + f_i(t)] T_i$$

$$\frac{dx_i(t)}{dt} = \lambda_i \cdot x_i(t) + f_i(t)$$

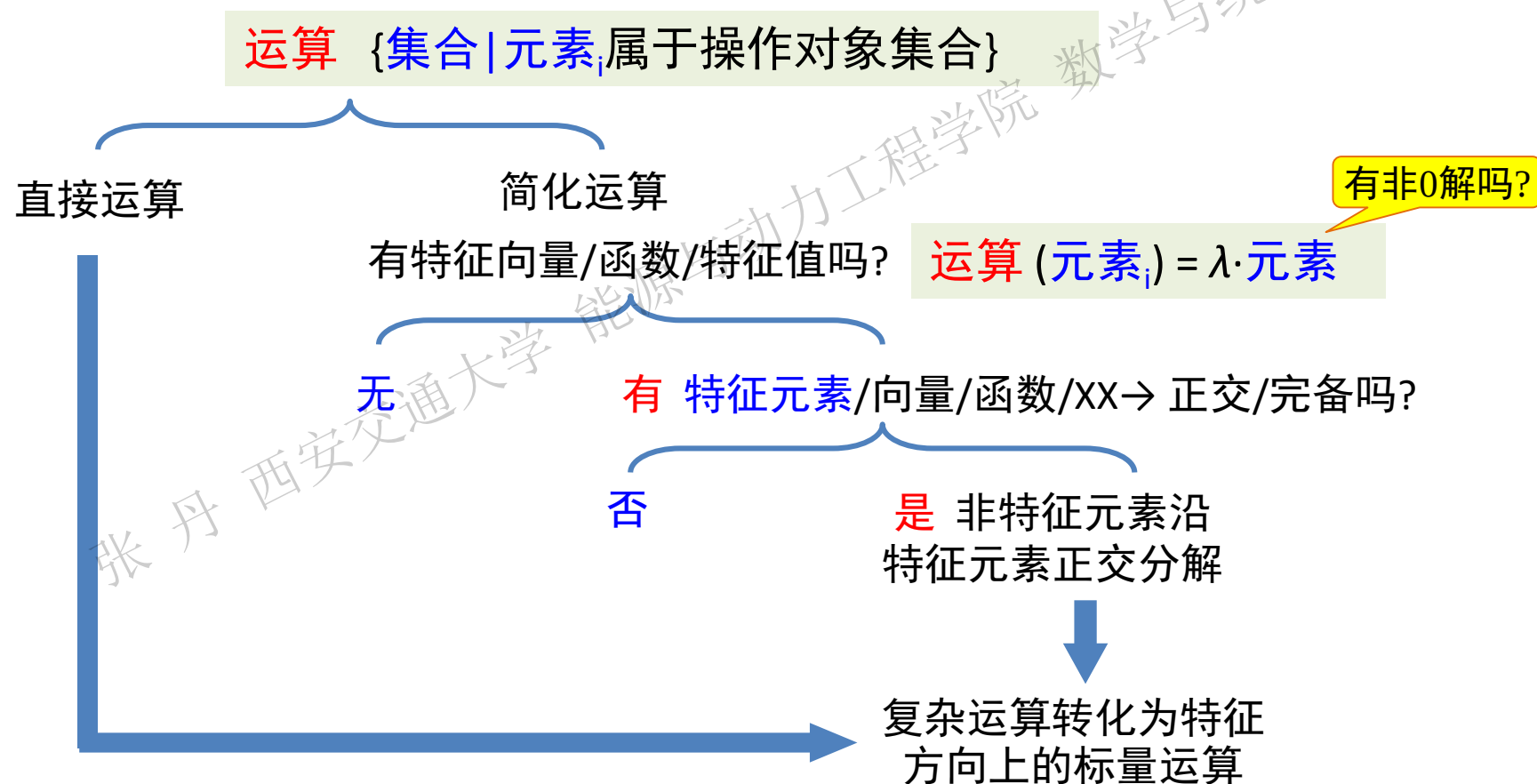
常数变易法

原方程的解为 $x(t) = \sum_{i=1}^n x_i(t) T_i, \quad x_i(t) = x_i^0 e^{\lambda_i t} + \int_0^t f_i(\tau) \cdot e^{\lambda_i(t-\tau)} d\tau$

2.1 特征值问题

小结

1. 特征值/特征向量(函数/方向)是针对运算效能的**反向衡量法**
2. 利用运算无可奈何的**特征元素 (向量/函数/方向)**可简化运算
3. 该简化运算的 思想/方法 不限于矩阵运算



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$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 波动/导热/Poisson 未知函数空间变化率一侧相同

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial \tau^2} - a^2 \frac{\partial^2 u}{\partial x^2} &= f \\ \frac{\partial u}{\partial \tau} - a^2 \frac{\partial^2 u}{\partial x^2} &= f \\ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= f \end{aligned} \right\}$$

定义二阶微分算子

$$A = -\frac{d^2}{dx^2}$$

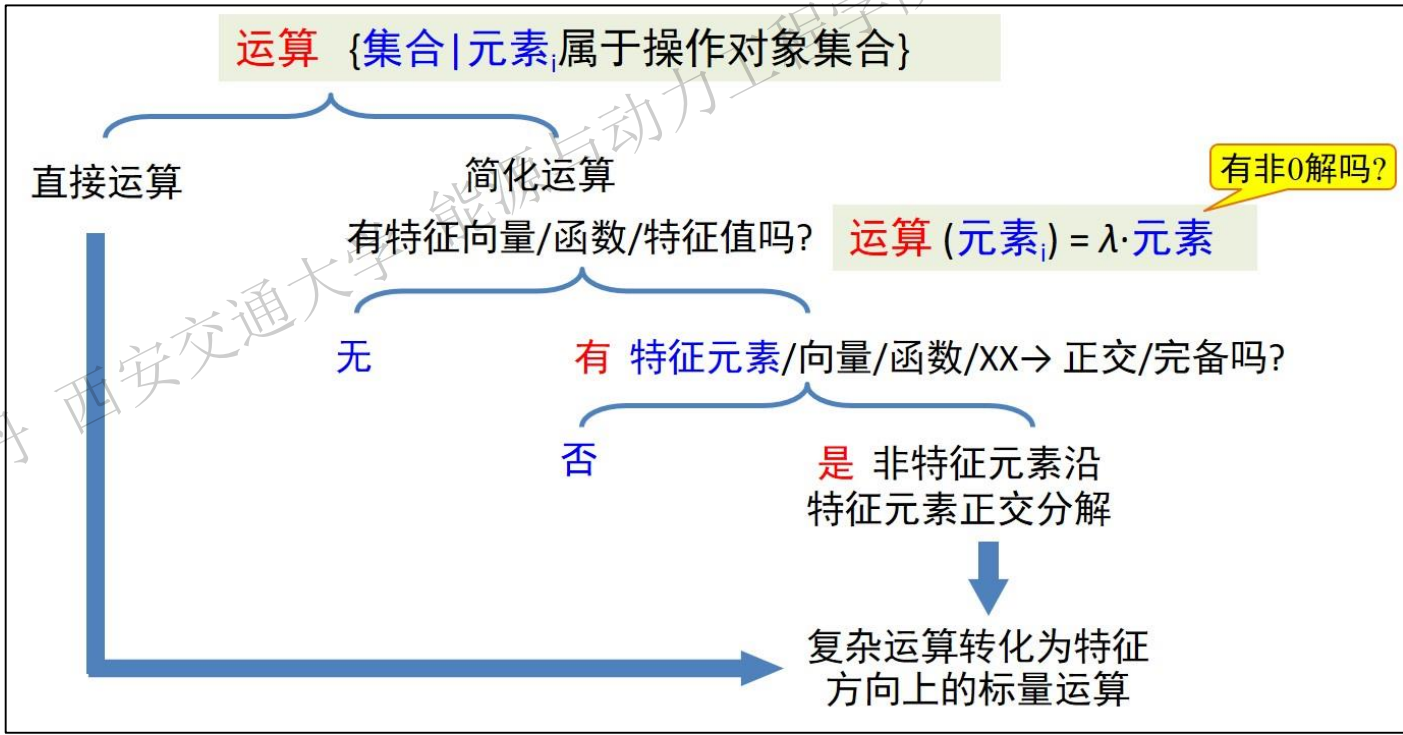
-Δ运算的一维形式

对象集合

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

第1/2/3类均可

数学意义 作用在函数空间 H 上的线性变换



2.2 分离变量法

预备知识1 函数正交性/正交函数系

函数正交 若函数 $f_1(x), f_2(x)$ 满足 $\int_a^b f_1(x)f_2(x)dx=0$, 称函数 $f_1(x), f_2(x)$ 在区间 $[a, b]$ 上正交

① 什么区间

② 什么函数

正交函数系 若定义在区间 $[a, b]$ 上的函数列 $\{f_1(x), f_2(x), f_2(x), \dots, f_n(x), \dots\}$ 中,

任意两个函数满足 $\int_a^b f_i(x)f_j(x)dx \begin{cases} =0, & (i \neq j) \\ \neq 0, & (i = j) \end{cases}$, 则称该函数系为正交

函数系

③ 平方模是多少

没0/函数名不同/系数不同

三角函数系 $\{1, \cos x, \cos 2x, \cos nx, \dots, \sin x, \sin 2x, \sin nx, \dots\}$ 在区间 $[-\pi, \pi]$ 上为正交函数系

平方模

不同函数定积分为0

相同函数平方模不同

$$\int_{-\pi}^{\pi} 1 \cdot dx = 2\pi$$

$$\int_{-\pi}^{\pi} \sin^2 nx \cdot dx = \left[\frac{1}{2n} (nx - \sin nx \cos nx) \right]_{-\pi}^{\pi} = \left[\frac{x}{2} - \frac{1}{4n} \sin 2nx \right]_{-\pi}^{\pi} = \pi$$

$$\int_{-\pi}^{\pi} \cos^2 nx \cdot dx = \left[\frac{1}{2n} (nx + \sin nx \cos nx) \right]_{-\pi}^{\pi} = \left[\frac{x}{2} + \frac{1}{4n} \sin 2nx \right]_{-\pi}^{\pi} = \pi$$

正交区间可推广至[整数 π , 整数 $\pi+2\pi$]

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值

$$A = -\frac{d^2}{dx^2}$$

$$AX(x) = \lambda X(x), \quad X(x) \in H$$

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

特征值/特征函数 → 是否存在? 是正是负? 具体是多少? → 依赖于边界条件

算子 A 的特征值问题

= 带有齐次边界条件ODE定解问题

[1, 1] 算子在给定区间两端都是第一类齐次边界

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

ODE边值问题

设 $X(x)$ 是相应于 λ 的非零解

同乘方程两端 $X(x)X''(x) + \lambda X^2(x) = 0$

积分得 $\int_0^l X(x)X''(x)dx + \lambda \int_0^l X^2(x)dx = 0$

分部积分

$$X(x)X'(x)|_0^l - \int_0^l [X'(x)]^2 dx + \lambda \int_0^l X^2(x)dx = 0$$

= 0

移项有

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

算子 $A = -\frac{d^2}{dx^2}$ 的特征值非负 → 能否求出

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 1]

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

算子 A 的特征值问题
= 带有齐次边界条件 ODE 定解问题

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

可否解出具体值?

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$

根据边界条件得 $C_1 = C_2 = 0, X(x) = 0$

方程为零解, 所以 $\lambda = 0$ 不是特征值

特征函数要求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$

根据边界条件得 $C_1 = 0, C_2 \sin(\sqrt{\lambda}l) = 0 \quad \sqrt{\lambda} \cdot l = n\pi, n \geq 1$

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), n \geq 1$$

1. 特征值/特征函数系的表示包含 3 要素
2. 0 是否为特征值需要单独讨论
3. 特征值 λ 是否包含 0 与特征值序号 n 是否从 0 开始 没有必然联系

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 1]

特征值/特征函数系性质

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

$A = -\frac{d^2}{dx^2}$

特征值 非负/递增 $\lambda_n = \left(\frac{n\pi}{l}\right)^2, n \geq 1$

特征函数系 $\left\{ X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \mid n \geq 1 \right\}$

正交

$$\int_{-l}^l \sin\left(\frac{n\pi}{l}x\right) \sin\left(\frac{m\pi}{l}x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin(n\theta) \sin(m\theta) d\theta = \begin{cases} 0, & n \neq m \\ l, & n = m \end{cases}$$

$\theta = \frac{\pi}{l}x, x = \frac{l}{\pi}\theta$



J. Fourier
1768-1830
French Math.
Physical
Scientist

1811, Fourier级数

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n = 0, 1, 2, 3 \dots)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n = 1, 2, 3 \dots)$$

完备 任意分段光滑的函数可展开为Fourier正弦级数

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{l} \right)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n = 1, 2, 3 \dots)$$



$$A = -\frac{d^2 f(x)}{dx^2}$$

对 $f(x)$ 的二阶求导运算 $\rightarrow$ 将函数 $f(x)$ 按特征函数系展开 $\rightarrow$ 转化为对其展开后各项的比例放缩 $\rightarrow$ 算子 **A** 代表的线性变换可正交分解 $\rightarrow$ 运算大大简化

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

特征值如何简化计算 [1, 1]

$$AX(x) = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X(l) = 0 \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin\left(\frac{n\pi}{l}x\right), \quad n \geq 1$$

例1

$$H = \{X(x) \in C^2[0, l] \mid X(x) \text{ 齐次 } x=0, l\}$$

$$A[f(x)] = -\frac{d^2 f(x)}{dx^2}$$

按特征函数系展开

$$f(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$= -\frac{d^2 \left[\sum_{n=1}^{\infty} f_n X_n(x) \right]}{dx^2}$$

f_n 与 x 无关, 求导/求和可交换次序

$$= \sum_{n=1}^{\infty} \left\{ f_n \frac{-d^2 [X_n(x)]}{dx^2} \right\}$$

$$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$$

利用特征值定义化简

$$= \sum_{n=1}^{\infty} f_n \cdot \lambda_n \cdot X_n(x)$$

求导运算 → 转化为简单的系数计算

以上为[1,1]型 → 改变边界条件 → 相应特征函数系也会改变 → Sturm-Liouville定理

2.1 特征值问题

$A = -\frac{d^2}{dx^2}$

二阶微分算子的特征值 Sturm-Liouville定理

向不同类齐次边界组合下的推广 $H = \{X(x) \in C^2[0,l] \mid X(x) \text{ 齐次 } x=0,l\}$

$A = -\frac{d^2}{dx^2}$
 $AX(x) = \lambda X(x)$
 $\rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X^{<k>}(0) = X^{<m>}(l) = 0, & 0 \leq k, m \leq 1, k, m \in \mathbb{Z}^+ \end{cases}$



J. Sturm
1803-1855
French Math.



J. Liouville
1809-1882
French Math.

- [1, 1]
- [1, 2]
- [2, 1]
- [2, 2]
- [1, 3]
- [3, 1]
- [2, 3]
- [3, 2]
- [3, 3]

① 存在

② 特征值 非负/递增 $0 \leq \lambda_1 < \lambda_2 < \dots < \lambda_n < \dots \rightarrow \infty$

③ 正交 $\int_0^l X_n(x) X_m(x) dx = \begin{cases} 0, & n \neq m \\ \int_0^l X_n^2(x) dx, & n = m \end{cases}$ 平方模

完备 在区间 $[0,l]$ 上, 任意分段光滑的函数可按特征函数系展开为广义Fourier级数

特征函数系 $\{X_n(x) \mid n \geq 1\}$

$$f(x) = \sum_{n=1}^{\infty} [f_n \cdot X_n(x)]$$
$$f_n = \frac{\int_0^l f(x) \cdot X_n(x) \cdot dx}{\int_0^l X_n^2(x) \cdot dx}, \quad (n=1,2,3\cdots)$$

任意齐次边界 $\rightarrow H$ 内任意函数可用特征函数系表示 $\rightarrow$ 均可简化为特征方向标量运算

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [1, 2]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X'(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$

根据边界条件得 $C_1 = C_2 = 0$, $X(x) = 0$

方程为零解, 所以 $\lambda = 0$ 不是特征值

特征函数要求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}x)$$

由边界条件 $C_1 = 0$, $C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}l) = 0$,

$$\sqrt{\lambda} \cdot l = \left(n + \frac{1}{2}\right)\pi = \frac{2n+1}{2}\pi, \quad n \geq 0$$

特征值/特征函数

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, \quad X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$$

$(n = 0, 1, 2, 3, \dots)$

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [2, 1]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = X(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$

根据边界条件得 $C_1 = C_2 = 0$, $X(x) = 0$

方程为零解, 所以 $\lambda = 0$ 不是特征值

特征函数要求非零解

当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda}x) + C_2 \sin(\sqrt{\lambda}x)$

$$X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda}x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda}x)$$

由边界条件 $C_2 = 0$, $C_1 \cos(\sqrt{\lambda}l) = 0$,

$$\sqrt{\lambda} \cdot l = \left(n + \frac{1}{2}\right)\pi = \frac{2n+1}{2}\pi, \quad n \geq 0$$

特征值/特征函数 $\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2$, $X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$
($n = 0, 1, 2, 3, \dots$)

$$A = -\frac{d^2}{dx^2}$$

2.1 特征值问题

二阶微分算子的特征值 [2, 2]

$$AX(x) = \lambda X(x) \rightarrow -\frac{d^2 X(x)}{dx^2} = \lambda X(x) \rightarrow \begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X'(0) = X'(l) = 0 \end{cases}$$

算子 A 的特征值非负

$$\lambda = \frac{\int_0^l [X'(x)]^2 dx}{\int_0^l X^2(x) \cdot dx} \geq 0$$

- 当 $\lambda = 0$ 时, 方程的通解为 $X(x) = C_1 x + C_2$, $X'(x) = C_1$
 根据边界条件得 $C_1 = 0$, $X(x) = C_2 \neq 0$
 方程为 **非零解**, 所以 $\lambda = 0$ 是特征值, 通常取 $X(x) = 1$
- 当 $\lambda > 0$ 时, 方程的通解为 $X(x) = C_1 \cos(\sqrt{\lambda} x) + C_2 \sin(\sqrt{\lambda} x)$
 $X'(x) = -C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} x) + C_2 \sqrt{\lambda} \cos(\sqrt{\lambda} x)$
 由边界条件 $C_2 = 0$, $C_1 \sqrt{\lambda} \sin(\sqrt{\lambda} l) = 0$, $\sqrt{\lambda} \cdot l = n\pi$, $n > 0$

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x\right), (n = 0, 1, 2, 3 \dots)$$

兼容 $\lambda=0$

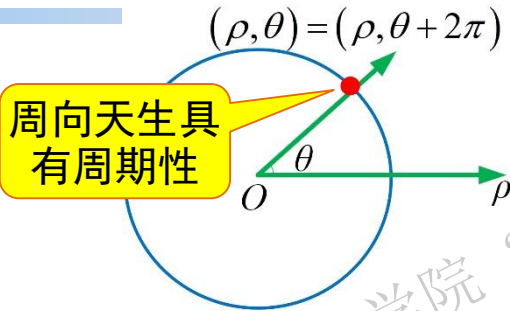
2.1 特征值问题

极坐标定解问题→周向/径向分离变量→周向存在特征值

$A = -\frac{d^2}{dx^2}$

二阶微分算子的特征值 周期性边界

$A = -\frac{d^2}{dx^2}$
 $A\Phi(x) = \lambda\Phi(x) \rightarrow \begin{cases} \Phi''(\theta) + \lambda\Phi(\theta) = 0, & 0 \leq \theta \leq 2\pi \\ \Phi(0) = \Phi(2\pi), \Phi'(0) = \Phi'(2\pi) \end{cases}$



当 $\lambda = 0$ 时, 通解为 $\Phi(\theta) = C_1\theta + C_2, \Phi'(\theta) = C_1$ 由边界条件 $C_1 = 0, \Phi(\theta) = C_2 \neq 0$

方程为**非零解** 所以 $\lambda = 0$ **是**特征值, 特征函数可取 $\Phi(\theta) = 1$

当 $\lambda > 0$ 时, 方程的通解为 $\Phi(\theta) = C_1 \cos(\sqrt{\lambda}\theta) + C_2 \sin(\sqrt{\lambda}\theta)$

$\Phi'(x) = -C_1\sqrt{\lambda} \sin(\sqrt{\lambda}\theta) + C_2\sqrt{\lambda} \cos(\sqrt{\lambda}\theta)$ 由边界条件

$$\begin{cases} C_1 = C_1 \cos(2\pi\sqrt{\lambda}) + C_2 \sin(2\pi\sqrt{\lambda}) \\ C_2\sqrt{\lambda} = -C_1\sqrt{\lambda} \sin(2\pi\sqrt{\lambda}) + C_2\sqrt{\lambda} \cos(2\pi\sqrt{\lambda}) \end{cases} \rightarrow \begin{bmatrix} 1 - \cos(2\pi\sqrt{\lambda}) & -\sin(2\pi\sqrt{\lambda}) \\ \sin(2\pi\sqrt{\lambda}) & 1 - \cos(2\pi\sqrt{\lambda}) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

不能同时为0

系数行列式=0, $1 - \cos(2\pi\sqrt{\lambda}) = 0$, 解得

$\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, n = 1, 2, 3 \dots$

2个特征函数

特征值/特征函数

$\lambda_0 = 0, \Phi_0(\theta) = 1, n = 0$
 $\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, n = 1, 2, 3 \dots$

若视 $\Phi_0(\theta) = 1 = \cos(0 \cdot \theta)$, 则 $n=0$ 与 $n = 1, 2, 3 \dots$ 可合并→但三角函数系中**1**与**sin/cos**平方模不同→展开极易出错→最好单列

2.1 特征值问题

小结

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数



齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0,1,2,3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = \{\cos n\theta, \sin n\theta\}, (n=1,2,3\cdots)$

2.1 特征值问题

例1 特征值问题 $\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < l \\ X(0) = X'(l) = 0 \end{cases}$

的特征值和特征函数是

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3\cdots)$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$$

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0, 1, 2, 3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0, 1, 2, 3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3\cdots)$

2.1 特征值问题

例2 求特征值问题 $\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < \pi \\ X'(0) = X(\pi) = 0 \end{cases}$

$$\lambda_n = \frac{(2n+1)^2}{4}, X_n(x) = \cos\left[\frac{(2n+1)}{2}x\right], (n = 0, 1, 2, 3 \cdots)$$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n = 1, 2, 3 \cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3 \cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n = 0, 1, 2, 3 \cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n = 0, 1, 2, 3 \cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n = 0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n = 1, 2, 3 \cdots)$

2.1 特征值问题

研究特征值问题的目的→将给定函数按特征函数系正交分解

特征函数系 平方模

将函数按 A 算子特征函数正交分解
→Fourier展开的变体→平方模是关键

平方模

$$\int_{-l}^l 1^2 \cdot dx = 2l$$
$$\theta = \frac{\pi}{l} x, x = \frac{l}{\pi} \theta$$
$$\int_{-l}^l \sin^2\left(\frac{n\pi}{l} x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin^2(n\theta) d\theta = l$$
$$\int_{-l}^l \cos^2\left(\frac{n\pi}{l} x\right) dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \cos^2(n\theta) d\theta = l$$
$$\int_{-l}^l \sin^2\left[\frac{(2n+1)\pi}{2l} x\right] dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \sin^2\left[\frac{(2n+1)}{2} \cdot \theta\right] d\theta = \frac{l}{\pi} \left[\frac{\theta}{2} - \frac{1}{4 \cdot \frac{(2n+1)}{2}} \sin\left(2 \frac{(2n+1)}{2} \theta\right) \right]_{-\pi}^{\pi} = l$$
$$\int_{-l}^l \cos^2\left[\frac{(2n+1)\pi}{2l} x\right] dx = \frac{l}{\pi} \int_{-\pi}^{\pi} \cos^2\left[\frac{(2n+1)}{2} \cdot \theta\right] d\theta = \frac{l}{\pi} \left[\frac{\theta}{2} + \frac{1}{4 \cdot \frac{(2n+1)}{2}} \sin\left(2 \frac{(2n+1)}{2} \theta\right) \right]_{-\pi}^{\pi} = l$$

A 算子特征函数正交区间是[-l, l]
平方模仅有1函数是 2l, 其余特征函数均为 l

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数		
齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x\right), (n=1, 2, 3 \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l} x\right], (n=0, 1, 2, 3 \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l} x\right], (n=0, 1, 2, 3 \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l} x\right), (n=0, 1, 2, 3 \dots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1, 2, 3 \dots)$

2.1 特征值问题

按特征函数系正交分解



间接阐明了三角函数的完备性

函数在 $[-\pi, \pi]$ 上:
1829, Dirichlet条件
1. 分段单调;
2. 除有限个第一类间断点外都连续,
则该函数的Fourier级数在 $[-\pi, \pi]$ 上收敛

1822, 定义在 $[0, l]$ 上函数Fourier展开

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$
$$a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} \cdot dx, \quad (n=0, 1, 2, 3 \dots)$$
$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3 \dots)$$

利用正交函数系将函数展开→正交分解

$$f(x) = \sum_{n=0}^{\infty} f_n X_n(x) = f_0 \cdot 1 + f_1 \cdot X_1(x) + \dots + f_n \cdot X_n(x) + \dots$$

n是特征值序号起点

展开系数计算通式

$$f_n = \frac{1}{\int_a^b X_n^2(x) dx} \int_a^b f(x) \cdot X_n(x) dx$$

$$f(x) = \sum_{n=1}^{\infty} f_n \cdot \sin\left(\frac{n\pi}{l} x\right) \Rightarrow f_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi}{l} x\right) dx = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l} x\right) dx$$

$$f(x) = \sum_{n=0}^{\infty} f_n \cdot \cos\left(\frac{n\pi}{l} x\right) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n \cdot \cos\left(\frac{n\pi}{l} x\right)$$

积分区间对折

$$f_0 = \frac{1}{2l} \int_{-l}^l f(x) \cdot 1 \cdot dx$$
$$= \frac{2}{2l} \int_0^l f(x) dx = \frac{1}{l} \int_0^l f(x) dx$$

$$f_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi}{l} x\right) dx = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l} x\right) dx$$

快速计算手册

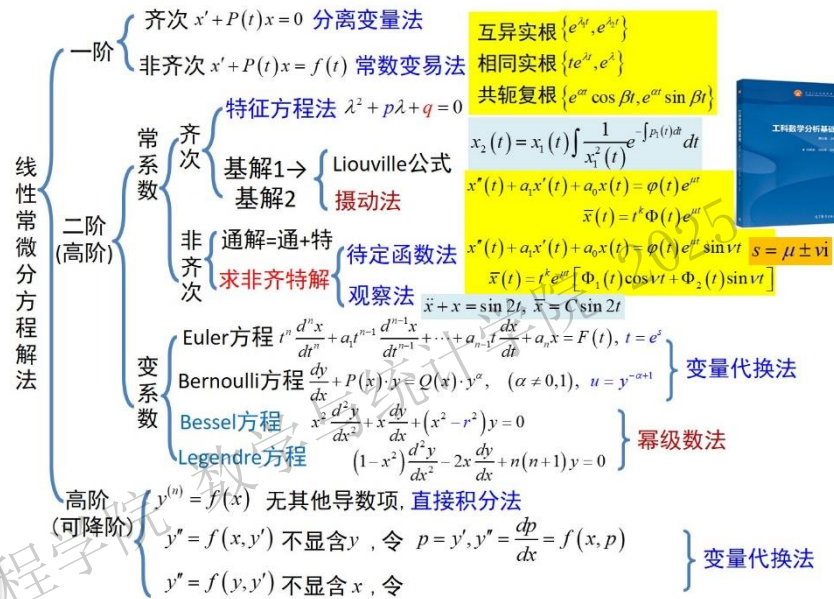
边界条件齐次化常用辅助函数 $u(x,t) = v(x,t) + w(x,t)$

边界类型	边界条件	辅助函数
[1,1]	$u(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{l}x + g_1(t)$
[2,1]	$u_x(0,t) = g_1(t), u(l,t) = g_2(t)$	$w(x,t) = g_1(t)(x-l) + g_2(t)$
[1,2]	$u(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = g_2(t)x + g_1(t)$
[2,2]	$u_x(0,t) = g_1(t), u_x(l,t) = g_2(t)$	$w(x,t) = \frac{g_2(t) - g_1(t)}{2l}x^2 + g_1(t)x$

算子 $A = -\frac{d^2}{dx^2}$ 不同齐次边界条件下的特征值/特征函数

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1,2,3\cdots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right], (n=0,1,2,3\cdots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right), (n=0,1,2,3\cdots)$
周期	$\begin{cases} \Phi''(\theta) + \lambda \Phi(\theta) = 0 \\ \Phi(0) = \Phi(2\pi), \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$	$\lambda_0 = 0, \Phi_0(\theta) = 1, (n=0)$ $\lambda_n = n^2, \Phi_n(\theta) = C_1 \cos n\theta + C_2 \sin n\theta, (n=1,2,3\cdots)$

线性ODE的解法 $x^{(n)} + P_1 x^{(n-1)} + P_2 x^{(n-2)} + \cdots + P_{n-1} x^{(1)} + P_n x^{(0)} = F$



数学物理方程

1. 数学建模和基本原理介绍

2. 分离变量法

3. Bessel函数

4. 积分变换法

5. Green函数法

6. 特征线法

7. Legendre多项式

2.1 特征值问题

2.2 分离变量法

2.3 基于MATLAB的定解问题求解方法

弦振动方程定解问题

热传导方程定解问题

平面位势方程边值问题

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例1

求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐1齐

解 第1步 分离变量 根据齐次方程/齐次边界, 假设 $u(x, t) = X(x)T(t)$

代入齐次波动方程有 $X(x)T''(t) - a^2 X''(x)T(t) = 0$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

分离常数

左右只可能同等于某个常数 $-\lambda$

边值问题初值问题

化为两个ODE $X''(x) + \lambda X(x) = 0$ $T''(t) + \lambda a^2 T(t) = 0$

由边界条件 $X(0)T(t) = X(l)T(t) = 0$ $T(t)$ 不恒为0

$$\begin{cases} X(0) = X(l) = 0 \\ X''(x) + \lambda X(x) = 0 \end{cases}$$

特征值/特征函数

$$A = -\frac{d^2}{dx^2}$$
$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) (n=1, 2, 3, \dots)$
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right] (n=0, 1, 2, 3, \dots)$
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right] (n=0, 1, 2, 3, \dots)$
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right) (n=0, 1, 2, 3, \dots)$

找出特征值/特征函数
→边值问题得解→
获得分离常数

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例1 求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/齐次

1齐

1齐

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

第2步 求解初值问题 将特征值 λ_n 代入初值ODE $T''(t) + \lambda a^2 T(t) = 0$

得到初值问题通解 $T_n(t) = C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t)$

每个 λ_n 对应的解 $u_n(x, t) = T_n(t) X_n(x) = \left[C_{1n} \sin(a\sqrt{\lambda_n}t) + C_{2n} \cos(a\sqrt{\lambda_n}t) \right] \sin\left(\frac{n\pi}{l}x\right)$

所有 λ_n 对应的解 叠加→形式解

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left[C_{1n} \sin\left(\frac{n\pi a}{l}t\right) + C_{2n} \cos\left(\frac{n\pi a}{l}t\right) \right] \sin\left(\frac{n\pi}{l}x\right)$$

应满足初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} C_{2n} \sin\left(\frac{n\pi}{l}x\right) = \varphi(x)$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} C_{1n} \frac{n\pi a}{l} \cdot \sin\left(\frac{n\pi}{l}x\right) = \psi(x)$$

系数不能吸收

初始条件按特征函数系正交分解

$$C_{2n} = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx, (n=1, 2, 3, \dots)$$

$$C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx, (n=1, 2, 3, \dots)$$

此 x 仅是积分变量

累加包括

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

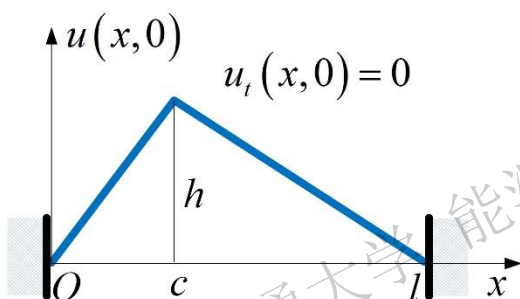
例1 求解一维波动定解问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & 0 < x < l, t > 0 & \text{波动/一维/齐次} \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 & \text{1齐} \quad \text{1齐} \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} \left[C_{1n} \sin\left(\frac{n\pi a}{l} t\right) + C_{2n} \cos\left(\frac{n\pi a}{l} t\right) \right] \sin\left(\frac{n\pi}{l} x\right)$$

$$C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx, \quad C_{2n} = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx, \quad (n=1, 2, 3, \dots)$$

讨论 两端固定/初速度为0的弦振动

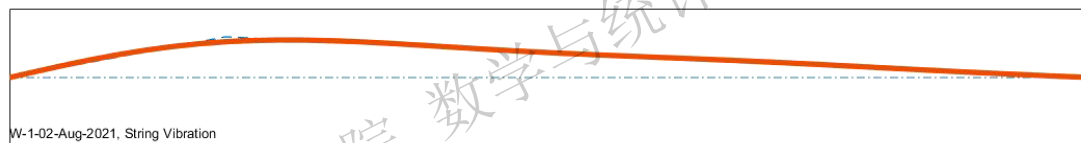


根据初始速度为0

$$\psi(x) = 0, \quad C_{1n} = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx = 0$$

初始位移可表示为

$$u(x, 0) = \varphi(x) = \begin{cases} \frac{h}{c} x, & 0 < x \leq c \\ \frac{-hx + hl}{l - c}, & c \leq x < l \end{cases}$$



$$\begin{aligned} C_{2n} &= \frac{2}{l} \left[\int_0^c \varphi(x) \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{2}{l} \left[\int_0^c \frac{h}{c} x \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \frac{-hx + hl}{l - c} \sin \frac{n\pi x}{l} \cdot dx \right] \\ &= \frac{2hl^2}{\pi^2 c(l - c)} \frac{1}{n^2} \sin \frac{n\pi}{l} c \end{aligned}$$

该定解问题的解为

$$u(x, \tau) = \frac{2hl^2}{\pi^2 c(l - c)} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cos\left(\frac{n\pi a}{l} \tau\right) \sin\left(\frac{n\pi}{l} x\right)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

分离变量的目标与推广

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f$$

$f = 0$

齐次PDE定解问题→可直接分离变量求解

分离变量法: PDE/ODE有质的不同

变量分离目标

1. 分离成独立函数 $u(x, t) = X(x)T(t)$
2. 找出各函数独立满足的ODE
3. 得出各ODE独立的定解条件

时/空解耦
的多个ODE
定解问题

若 $u(x, t) = X(x) + T(t)$ { 不同量纲不能相加
无法找出各自ODE独立定解条件

齐次PDE的求解并未体现特征值问题简化计算的功能

$f \neq 0$

无法直接分离变量→但特征函数系 $\{X_n(x)\}$ 正交/完备→正交分解
 $u(x, t) = \sum_{n=0,1}^{\infty} X_n(x)T_n(t) \rightarrow T_n(t)$ 待定函数法利用初始条件确定→特征函数法/正交分解法/广义分离变量法

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1齐

解 边界已经齐次

第1步 特征值问题, 设 $u(x, t) = X(x)T(t)$

代入对应的齐次方程

$$X(x)T''(t) - a^2 X''(x)T(t) = 0$$

$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$T''(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$
$$\left. \begin{array}{l} \text{齐次边界 } T(t) \text{ 不恒为 } 0 \\ X(0) = X(l) = 0 \end{array} \right\}$$

边界条件由边值问题实现

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3, \dots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3, \dots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3, \dots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3, \dots$)

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right), (n=1, 2, 3, \dots)$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

特征值/特征函数

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \\ (n=1, 2, 3, \dots)$$

第2步 正交分解

待定函数

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l}x\right)$$

自由项/初始条件按特征函数系展开

$$A = \sum_{n=1}^{\infty} f_n X_n(x) = \sum_{n=1}^{\infty} f_n \sin\left(\frac{n\pi}{l}x\right)$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$\psi(x) = \sum_{n=1}^{\infty} \psi_n X_n(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

$$f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l}x\right) dx = \frac{2A}{n\pi} [1 - \cos n\pi]$$

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$\psi_n = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

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波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right) \\ (n=1, 2, 3, \dots)$$

$$A = \sum_{n=1}^{\infty} f_n X_n(x),$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l} x \right), \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l} x \right)$$

第3步 建立初值问题ODE $T_n(t)$

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x) = \sum_{n=1}^{\infty} T_n(t) \sin\left(\frac{n\pi}{l} x \right)$$

假设可逐项求导

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) (-X_n''(x)) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$X_n(x)$ 是 A 运算特征函数 $X_n''(x) + \lambda X_n(x) = 0$

特征值简化运算

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) [\lambda_n X_n(x)] = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

比较系数 $T_n''(t) + a^2 \lambda_n T_n(t) = f_n$

用原PDE初始条件 $\rightarrow$ 初值问题ODE初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = \varphi(x) = \sum_{n=1}^{\infty} \varphi_n X_n(x)$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} T_n'(0) X_n(x) = \psi(x) = \sum_{n=1}^{\infty} \psi_n X_n(x)$$

建立了初值问题ODE $T_n(t)$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

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波动/一维/非齐

1齐1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$$
$$(n=1, 2, 3\cdots)$$

$$A = \sum_{n=1}^{\infty} f_n X_n(x),$$

$$\varphi(x) = \sum_{n=1}^{\infty} \varphi_n \sin\left(\frac{n\pi}{l}x\right), \psi(x) = \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi}{l}x\right)$$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = \varphi_n, T_n'(0) = \psi_n \end{cases}$$

二阶线性非齐次ODE定解问题

通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

f_n 为常数, 故 $\bar{T}_n$ 也为常数

$$T_n'(t) = -C_{1n} a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n} a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t)$$
$$T_n''(t) = -C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) - C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t)$$

$$T_n''(t) + a^2 \lambda_n T_n(t) = f_n$$
$$= -C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) - C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t)$$
$$+ C_{1n} a^2 \lambda_n \cos(a\sqrt{\lambda_n}t) + C_{2n} a^2 \lambda_n \sin(a\sqrt{\lambda_n}t)$$
$$+ a^2 \lambda_n \bar{T}_n = f_n$$

$$\bar{T}_n = \frac{f_n}{a^2 \lambda_n}$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{f_n}{a^2 \lambda_n}$$

根据初始条件确定系数

$$\begin{cases} T_n(0) = C_{1n} + \frac{f_n}{a^2 \lambda_n} = \varphi_n \\ T_n'(0) = C_{2n} a\sqrt{\lambda_n} = \psi_n \end{cases}$$

$$\begin{cases} C_{1n} = \varphi_n - \frac{f_n}{a^2 \lambda_n} \\ C_{2n} = \frac{l}{n\pi a} \psi_n \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

1齐

$$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin\left(\frac{n\pi}{l} x \right) \\ (n=1, 2, 3, \dots)$$

初值解代入得出原定解问题的解

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x)$$

$$= \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{f_n}{a^2 \lambda_n} \right) \cos(a \sqrt{\lambda_n} t) + \frac{l \cdot \psi_n}{n \pi a} \sin(a \sqrt{\lambda_n} t) + \frac{f_n}{a^2 \lambda_n} \right] \sin\left(\frac{n\pi}{l} x \right)$$

$$= \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{l^2}{n^2 \pi^2 a^2} f_n \right) \cos \frac{n\pi a}{l} t + \frac{l}{n \pi a} \psi_n \cdot \sin \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} f_n \right] \sin \frac{n\pi}{l} x$$

$$C_{1n} = \varphi_n - \frac{f_n}{a^2 \lambda_n}, C_{2n} = \frac{l}{n \pi a} \psi_n \\ T_n(t) = C_{1n} \cos(a \sqrt{\lambda_n} t) + C_{2n} \sin(a \sqrt{\lambda_n} t) + \frac{f_n}{a^2 \lambda_n}$$

$$f_n = \frac{2}{l} \int_0^l A \sin\left(\frac{n\pi}{l} x \right) dx = \frac{2A}{n\pi} (1 - \cos n\pi)$$

其中

$$\varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi}{l} x \right) dx, \psi_n = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{n\pi}{l} x \right) dx$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

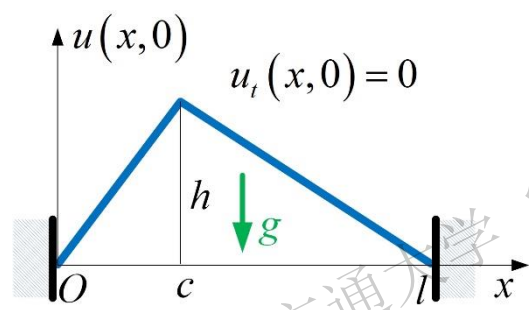
$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \quad (n=1, 2, 3\cdots)$$

讨论 考虑重力的弦振动



根据初始速度为0
 $\psi(x)=0, \psi_n = \frac{2}{n\pi a} \int_0^l \psi(x) \sin \frac{n\pi x}{l} \cdot dx = 0$

初始位移可表示为
$$u(x, 0) = \varphi(x) = \begin{cases} \frac{h}{c}x, & 0 < x \leq c \\ \frac{-hx + hl}{l - c}, & c \leq x < l \end{cases}$$

$$u(x, t) = \sum_{n=1}^{\infty} \left[\left(\varphi_n - \frac{l^2}{n^2 \pi^2 a^2} f_n \right) \cos \frac{n\pi a}{l} t + \frac{l}{n\pi a} \psi_n \cdot \sin \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} f_n \right] \sin \frac{n\pi}{l} x$$
$$f_n = \frac{2A}{n\pi} [1 - \cos n\pi], \varphi_n = \frac{2}{l} \int_0^l \varphi(x) \sin \left(\frac{n\pi}{l} x \right) dx, \psi_n = \frac{2}{l} \int_0^l \psi(x) \sin \left(\frac{n\pi}{l} x \right) dx$$

$$\varphi_n = \frac{2}{l} \left[\int_0^c \varphi(x) \sin \frac{n\pi x}{l} \cdot dx + \int_c^l \varphi(x) \sin \frac{n\pi x}{l} \cdot dx \right]$$
$$= \frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi}{l} c$$

$$u(x, t) = \sum_{n=1}^{\infty} \left[\varphi_n \cos \frac{n\pi a}{l} t + \frac{l^2}{n^2 \pi^2 a^2} \left(1 - \cos \frac{n\pi a}{l} t \right) f_n \right] \sin \frac{n\pi}{l} x$$
$$= \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t \right) \left(1 - (-1)^n \right) \right] \sin \frac{n\pi}{l} x$$

$\cos n\pi = (-1)^n$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例2 求解一维波动定解问题, 其中自由项 A 为常数

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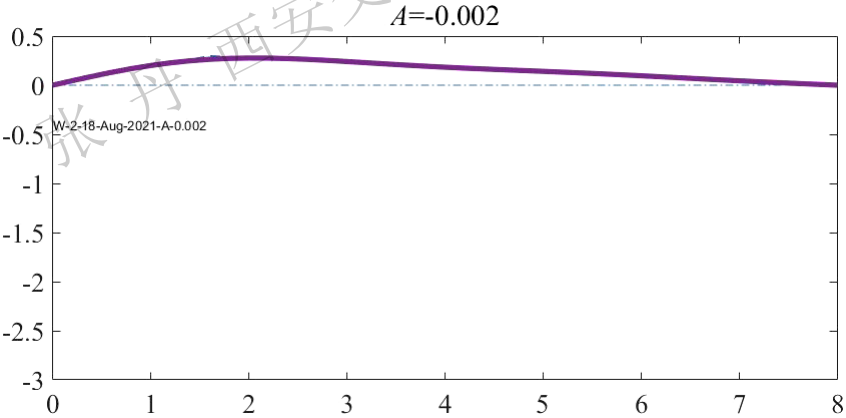
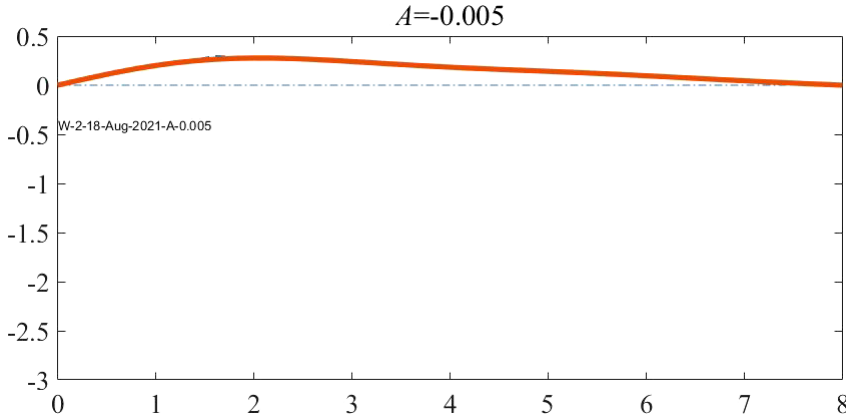
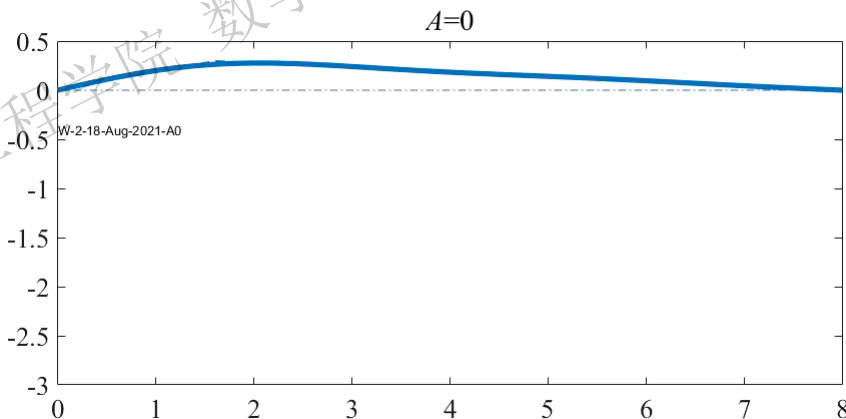
波动/一维/非齐

1齐1齐

$$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right) \quad (n=1, 2, 3 \dots)$$

讨论

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t\right) \left(1 - (-1)^n\right) \right] \sin \frac{n\pi}{l} x$$



$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

驻波法的由来

$$\begin{cases} u_{tt} - a^2 u_{xx} = A, & 0 < x < l, t > 0 \\ u(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x), & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

1齐

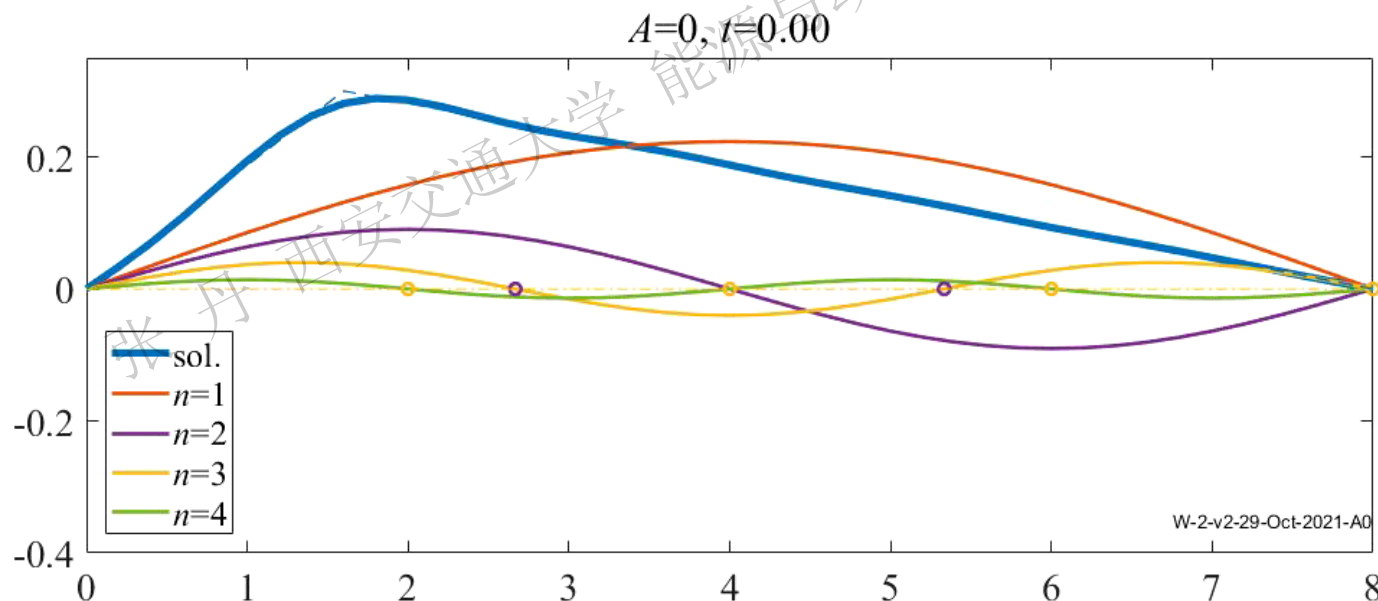
1齐

$$u(x, t) = \sum_{n=1}^{\infty} \left[\frac{2hl^2}{\pi^2 c(l-c)} \frac{1}{n^2} \sin \frac{n\pi c}{l} \cdot \cos \frac{n\pi a}{l} t + \frac{2Al^2}{\pi^3 a^2} \frac{1}{n^3} \left(1 - \cos \frac{n\pi a}{l} t \right) (1 - \cos n\pi) \right] \sin \frac{n\pi}{l} x$$

1. 解由无数简谐振动叠加

2. 各振动都有振幅始终为0的点 → 该振动称为驻波 → 分离变量法 = 驻波法 = 特征函数法

3. 驻波中振幅始终为0的点称为节点 $\sin \frac{n\pi}{l} x = 0, \frac{n\pi}{l} x = k\pi, x = k \frac{l}{n}, (1 \leq k \leq n-1)$



驻波的运用

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

预备知识 固有频率/共振 固有频率→与时间有关的固有属性→齐次PDE分离→ $T(t)$ 体现

$u_{tt} - a^2 u_{xx} = 0$ 分离变量 $u(x, t) = X(x)T(t)$

边界条件→特征值/特征函数系 $\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$

$X''(x) + \lambda X(x) = 0 \quad T''(t) + \lambda a^2 T(t) = 0$

无论何种齐次边界→初值通解形式一致→弦上任一质点都做周期性运动

$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t)$ 以[1,1]型边界为例

固有频率←特征值←边界类型 $\omega_n = a\sqrt{\lambda_n} = a \frac{n\pi}{l}, (n=1, 2, 3\cdots)$

$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$
 $\lambda = -\frac{1}{X(x)} \cdot \frac{d^2 X(x)}{dx^2}$
 $[a\sqrt{\lambda_n}] = \left[\frac{\text{m}}{\text{s}} \sqrt{\frac{1}{\text{m}^2}} \right] = \left[\frac{1}{\text{s}} \right] = [\text{Hz}]$
 $[\lambda_n] = \left[\frac{1}{\text{m}^2} \right]$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \sin\left(\frac{n\pi}{l}x\right)$ ($n=1, 2, 3\cdots$)
[1,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \sin\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3\cdots$)
[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l}\right]^2, X_n(x) = \cos\left[\frac{(2n+1)\pi}{2l}x\right]$ ($n=0, 1, 2, 3\cdots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l}\right)^2, X_n(x) = \cos\left(\frac{n\pi}{l}x\right)$ ($n=0, 1, 2, 3\cdots$)

最低阶固有频率→固有主频

$\omega_1 = a\sqrt{\lambda_1} = \frac{1 \cdot \pi a}{l}$
 $\omega_2 = a\sqrt{\lambda_2} = \frac{2 \cdot \pi a}{l}$
 $\vdots$
 $\omega_k = a\sqrt{\lambda_k} = \frac{k \cdot \pi a}{l}$
 $\vdots$
 $\omega_n = a\sqrt{\lambda_n} = \frac{n \cdot \pi a}{l}$

- 1.共振:周期性外力频率等于任一阶固有频率
- 2.外力一次只能等于某一个固有频率
- 3. k 阶共振本质是该阶初值问题特解不同

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

解 第1步 特征值问题, 设 $u(x, t) = X(x)T(t)$

代入对应的**齐次PDE**

$$X(x)T''(t) - a^2 X''(x)T(t) = 0$$
$$\frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

分离变量法精髓

$$T''(t) + \lambda a^2 T(t) = 0, X''(x) + \lambda X(x) = 0$$

齐次边界 $T(t)$ 不恒为0

$X'(0) = X(l) = 0$

特征值/特征函数

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right], (n = 0, 1, 2, 3 \cdots)$$

齐次边界类型	特征值问题	特征值/特征函数
[1,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \sin \left(\frac{n\pi}{l} x \right)$ ($n = 1, 2, 3 \cdots$)
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[2,1]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X(l) = 0 \end{cases}$	$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right]$ ($n = 0, 1, 2, 3 \cdots$)
[2,2]	$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(l) = 0 \end{cases}$	$\lambda_n = \left(\frac{n\pi}{l} \right)^2, X_n(x) = \cos \left(\frac{n\pi}{l} x \right)$ ($n = 0, 1, 2, 3 \cdots$)

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \\ (n = 0, 1, 2, 3 \dots)$$

第2步 正交分解

分离变量的形式解

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

自由项/初始条件 按特征函数系展开

$$\rho^{-1} \sin \omega t = \sum_{n=0}^{\infty} f_n X_n(x) = \sum_{n=0}^{\infty} f_n \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

$$f_n = \frac{2}{l} \int_0^l \rho^{-1} \sin \omega t \cdot \cos \left[\frac{(2n+1)\pi}{2l} x \right] dx = \frac{(-1)^n}{2n+1} \cdot \frac{4}{\pi \rho} \sin \omega t = k_n \sin \omega t$$

简化后续
表达式

定义在 $[0, l]$ 上函数 $f(x)$ 的Fourier展开/正交分解

$$f(x) = f_0 \cdot 1 + \sum_{n=1}^{\infty} f_n X_n(x)$$

$$f_0 = \frac{1}{l} \int_0^l f(x) \cdot 1 \cdot dx,$$

$$f_n = \frac{2}{l} \int_0^l f(x) \cdot X_n(x) dx$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第3步 建立初值ODE

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x) = \sum_{n=0}^{\infty} T_n(t) \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

假设可逐项求导

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) (-X_n''(x)) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$X_n(x) \text{ 是特征函数} \quad X_n''(x) + \lambda_n X_n(x) = 0$$

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) + a^2 \sum_{n=1}^{\infty} T_n(t) [\lambda_n X_n(x)] = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\sum_{n=1}^{\infty} [T_n''(t) + a^2 \lambda_n T_n(t)] X_n(x) = \sum_{n=1}^{\infty} f_n X_n(x)$$

$$\text{比较系数} \quad T_n''(t) + a^2 \lambda_n T_n(t) = f_n$$

用原PDE初始条件 $\rightarrow$ 初值ODE初始条件

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) X_n(x) = 0$$

特征函数非0

$$u_t(x, 0) = \sum_{n=1}^{\infty} T_n'(0) X_n(x) = 0$$

建立了初值问题ODE

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

2.2 分离变量法

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题

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波动/一维/非齐

2齐1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

$(n = 0, 1, 2, 3 \dots)$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0 \end{cases}$$

二阶ODE非齐次项含三角函数时特解取法

$$\ddot{x} + a_1 \dot{x} + a_2 x = e^{\mu t} \varphi(t) \sin vt$$

↓

$$\bar{x} = t^k e^{\mu t} [Z_1(t) \cos vt + Z_2(t) \sin vt]$$

$s^2 + a_1 s + a_2 = 0, \mu \pm iv$ $\left\{ \begin{array}{l} \text{非特征根 } k=0 \\ \text{是特征根 } k=1 \end{array} \right.$

$Z_1(t), Z_2(t)$ 是与 $\varphi(t)$ 同次幂实多项式



二阶线性非齐次ODE定解问题
通解=齐次通解+非齐特解

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

λ_n 不止一个 $\rightarrow \pm \omega i$ 是不是根? 是哪个根 $\rightarrow$ 结合物理意义 $\rightarrow$ 初值问题分类计算框架

特解 $\bar{T}_n$ 的取法需分类讨论

$\pm \omega i$ 是否为 $s^2 + a^2 \lambda_n = 0$ 的根

否 $\left\{ \begin{array}{l} \omega^2 \neq a^2 \lambda_n, \bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t \end{array} \right.$

是 $\left\{ \begin{array}{l} \omega^2 = a^2 \lambda_n = \omega_n^2, \bar{T}_n = t(\bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t) \end{array} \right.$

固有频率的平方

辅助等式可用于后续化简

2.2 分离变量法

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

一维波动定解问题 周期性外力作用下的弦振动, 左端为自由端, 右端固定, 外力密度 $\sin \omega t$

例3
$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐
2齐 1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3, \dots)$$

$$\begin{cases} T_n(0) = 0, T'_n(0) = 0 \\ T''_n(t) + a^2 \lambda_n T_n(t) = f_n = e^{0t} k_n \sin \omega t \end{cases}$$

二阶ODE非齐次项含三角函数时特解取法
 $\ddot{x} + a_1 \dot{x} + a_2 x = e^{\mu t} \varphi(t) \sin vt$
 $\bar{x} = t^k e^{\mu t} [Z_1(t) \cos vt + Z_2(t) \sin vt]$
 $s^2 + a_1 s + a_2 = 0, \mu \pm iv \begin{cases} \text{非特征根 } k=0 \\ \text{是特征根 } k=1 \end{cases}$
 $Z_1(t), Z_2(t)$ 是与 $\varphi(t)$ 同次幂实多项式

$$-\frac{d^2 X(x)}{dx^2} = \lambda X(x)$$
$$\lambda = -\frac{1}{X(x)} \cdot \frac{d^2 X(x)}{dx^2}$$

特征方程 $s^2 + a^2 \lambda_n = 0$
特征根 $s = \pm a\sqrt{\lambda_n} i = \pm \omega_n i$

$\mu + vi = 0 \pm \omega i$

$$[a\sqrt{\lambda_n}] = \left[\frac{m}{s} \sqrt{\frac{1}{m^2}} \right] = \left[\frac{1}{s} \right] = [Hz]$$

数学 $\omega = a\sqrt{\lambda_n}$ 物理 $\omega = \omega_n = a\sqrt{\lambda_n}$

固有频率/不止一个/n阶固有频率

是不是特征根?

$$\omega^2 = \omega_n^2 = a^2 \lambda_n$$

是不是固有频率?

辅助等式→后续共振情形解的化简

ω 不等于任意固有频率 $\omega^2 \neq \omega_n^2 = a^2 \lambda_n$
无共振, 所有初值问题特解相同

ω 等于k次固有频率 $\omega^2 = \omega_k^2 = a^2 \lambda_k$
有共振, 但仅k次谐波对应的特解与众不同

0 $\pm \omega i$ 非特征根 $\bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t$

$$\bar{T}_n = \begin{cases} \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t & (n \neq k) \text{ 非特征根} \\ t(\bar{C}_{1k} \cos \omega t + \bar{C}_{2k} \sin \omega t) & (n = k, \omega = a\sqrt{\lambda_k}) \text{ 是特征根} \end{cases}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n = 0, 1, 2, 3 \dots)$$

第4步 求初值问题

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \geq 0 \end{cases}$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \frac{k_n}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$T_n'(t) = -C_{1n} a\sqrt{\lambda_n} \sin(a\sqrt{\lambda_n}t) + C_{2n} a\sqrt{\lambda_n} \cos(a\sqrt{\lambda_n}t) - \frac{k_n \omega}{a^2 \lambda_n - \omega^2} \cos \omega t$$

$$T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$$

由初始条件 $C_{1n} = 0, C_{2n} = \frac{k_n \omega}{a\sqrt{\lambda_n}(\omega^2 - a^2 \lambda_n)}$

1 非共振 $\omega^2 \neq a^2 \lambda_n, \bar{T}_n = \bar{C}_{1n} \cos \omega t + \bar{C}_{2n} \sin \omega t$

$$\bar{T}_n' = -\bar{C}_{1n} \omega \sin \omega t + \bar{C}_{2n} \omega \cos \omega t$$

$$\bar{T}_n'' = -\bar{C}_{1n} \omega^2 \cos \omega t - \bar{C}_{2n} \omega^2 \sin \omega t$$

代入比较系数 $\bar{C}_{1n} = 0, \bar{C}_{2n} = \frac{k_n}{a^2 \lambda_n - \omega^2}$

$$\bar{T}_n = \frac{k_n}{a^2 \lambda_n - \omega^2} \sin \omega t$$

$$T_n(t) = \frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right)$$

$$u(x, t) = \sum_{n=0}^{\infty} T_n(t) X_n(x)$$

原定解问题非共振解

$$= \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐	
2齐	1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3 \dots)$$

第4步 求初值问题 $T_n(t) = C_{1n} \cos(a\sqrt{\lambda_n}t) + C_{2n} \sin(a\sqrt{\lambda_n}t) + \bar{T}_n$

2 共振

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \geq 0 \end{cases}$$

系统固有频率 ω_n 有多个, 外力频率 ω 一次只能等于一个谐波频率
不妨设 $\omega^2 = \omega_k^2 = a^2 \lambda_k, (n=k)$

非共振谐波 $n \neq k$
共振谐波 $n = k$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = k_n \sin \omega t \\ T_n(0) = 0, T_n'(0) = 0, n \neq k \end{cases}$$

与非共振相同

叠加原理

$$u(x, t) = T_k(t) X_k(x) + \sum_{n=0, n \neq k}^{\infty} T_n(t) X_n(x)$$

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

本例取 $n = k, \omega = a\sqrt{\lambda_k} = \frac{(2n+1)\pi a}{2l}$

特解形式 $\bar{T}_k = t(\bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t)$ 下面先确定特解系数

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3, \dots)$$

第4步 求初值问题 共振

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

$$\omega^2 = a^2 \lambda_k$$

$$\begin{aligned} T_k''(t) + a^2 \lambda_k T_k(t) &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega + a^2 \lambda_k \cdot t \bar{C}_1) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega + a^2 \lambda_k \cdot t \bar{C}_2) \sin \omega t \\ &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega + \omega^2 t \bar{C}_1) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega + \omega^2 t \bar{C}_2) \sin \omega t \\ &= (+2\bar{C}_2 \omega) \cos \omega t - (2\bar{C}_1 \omega + 2\omega^2 t \bar{C}_2) \sin \omega t = k_k \sin \omega t \end{aligned}$$

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) + \bar{T}_k$$

$$\bar{T}_k = t(\bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t)$$

$$\begin{aligned} \bar{T}_k' &= \bar{C}_1 \cos \omega t + \bar{C}_2 \sin \omega t \\ &\quad + t(-\bar{C}_1 \omega \sin \omega t + \bar{C}_2 \omega \cos \omega t) \end{aligned}$$

$$\begin{aligned} \bar{T}_k'' &= (-\bar{C}_1 \omega^2 t + 2\bar{C}_2 \omega) \cos \omega t \\ &\quad - (\bar{C}_2 \omega^2 t + 2\bar{C}_1 \omega) \sin \omega t \end{aligned}$$

代入比较系数 $\bar{C}_1 = -\frac{k_k}{2\omega}, \bar{C}_2 = 0, \bar{T}_k = -\frac{k_k t}{2\omega} \cos \omega t$

该初值问题的通解为

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) - \frac{k_k t}{2\omega} \cos \omega t$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

例3 周期性外力作用下的弦振动, $x=0$ 端为自由端, $x=l$ 端固定, 外力密度 $\sin \omega t$

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right] \quad (n=0, 1, 2, 3, \dots)$$

第4步 求初值问题 共振

共振发生时原定解问题的解

$$\begin{cases} T_k''(t) + a^2 \lambda_k T_k(t) = k_k \sin \omega t \\ T_k(0) = 0, T_k'(0) = 0, n = k \end{cases}$$

$$T_k(t) = C_1 \cos(a\sqrt{\lambda_k}t) + C_2 \sin(a\sqrt{\lambda_k}t) - \frac{k_k t}{2\omega} \cos \omega t$$

$$T_k'(t) = -C_1 a\sqrt{\lambda_k} \sin(a\sqrt{\lambda_k}t) + C_2 a\sqrt{\lambda_k} \cos(a\sqrt{\lambda_k}t) + \frac{k_k t}{2} \sin \omega t - \frac{k_k}{2\omega} \cos \omega t$$

由初始条件 $C_1 = 0, C_2 = \frac{k_k}{2\omega^2}$

$$T_k(t) = \frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t$$

初值问题解毕

$$\begin{aligned} u(x, t) &= \sum_{n=0}^{\infty} T_n(t) X_n(x) \\ &= T_k(t) X_k(x) + \sum_{n=0, n \neq k}^{\infty} T_n(t) X_n(x) \\ &= \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \left[\frac{(2k+1)\pi}{2l} x \right] \\ &\quad + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \cdot \left(\frac{\omega}{a\sqrt{\lambda_n}} \sin(a\sqrt{\lambda_n}t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x \end{aligned}$$

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = f(x, t)$$

2.2 分离变量法

一维波动定解问题

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$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

波动/一维/非齐

2齐

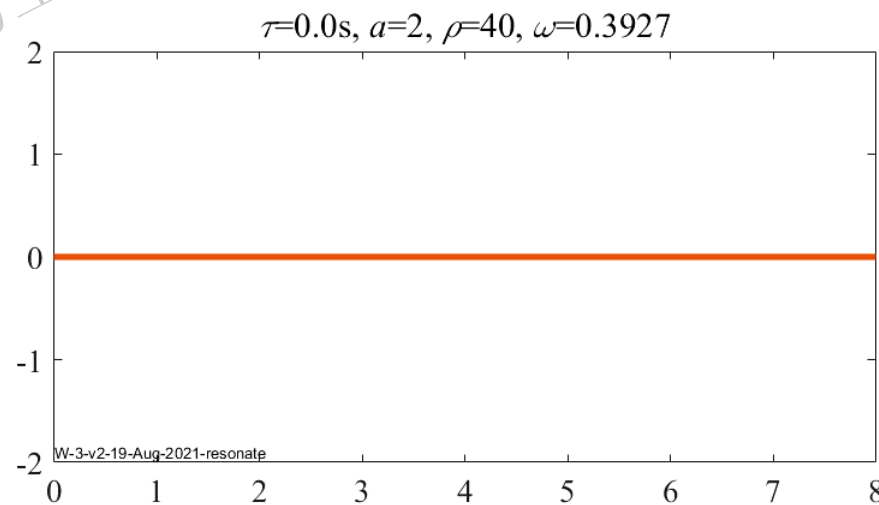
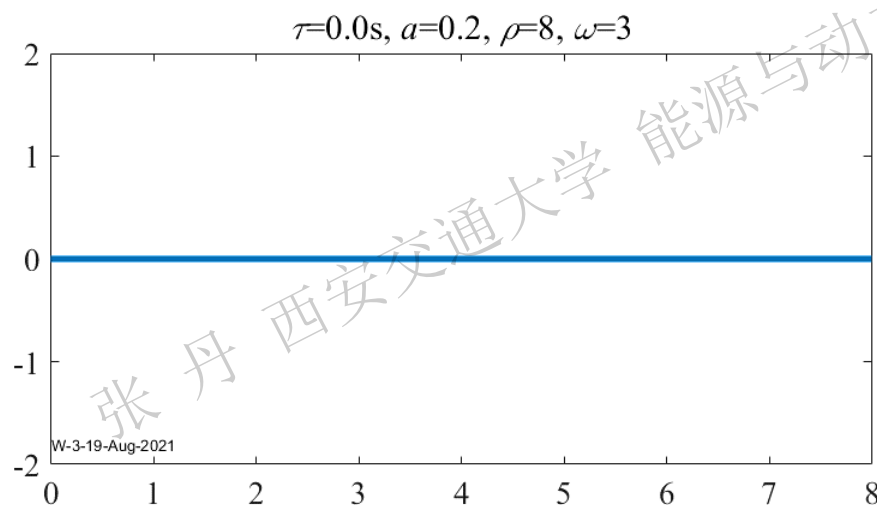
1齐

$$\lambda_n = \left[\frac{(2n+1)\pi}{2l} \right]^2, X_n(x) = \cos \left[\frac{(2n+1)\pi}{2l} x \right]$$

$$k_n = \frac{(-1)^n}{2n+1} \cdot \frac{4}{\pi \rho}, (n=0, 1, 2, 3, \dots)$$

讨论

$$u(x, t) = \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$



$$u(x, t) = \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \frac{\pi}{2l} x + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \left(\frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) - \sin \omega t \right) \right] \cos \frac{(2n+1)\pi}{2l} x$$

2.2 分离变量法

一维波动定解问题

例3

$$\begin{cases} u_{tt} - a^2 u_{xx} = \rho^{-1} \sin \omega t, & 0 < x < l, t > 0 \\ u_x(0, t) = 0, u(l, t) = 0, & t \geq 0 \\ u(x, 0) = 0, u_t(x, 0) = 0, & 0 \leq x \leq l \end{cases}$$

讨论

$$u(x, t) = \sum_{n=0}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \begin{pmatrix} \frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) \\ -\sin \omega t \end{pmatrix} \right] \cos \frac{(2n+1)\pi}{2l} x$$

$$u(x, t) = \left(\frac{k_k}{2\omega^2} \sin \omega t - \frac{k_k t}{2\omega} \cos \omega t \right) \cos \frac{\pi}{2l} x + \sum_{n=0, n \neq k}^{\infty} \left[\frac{k_n}{\omega^2 - a^2 \lambda_n} \begin{pmatrix} \frac{\omega}{a \sqrt{\lambda_n}} \sin(a \sqrt{\lambda_n} t) \\ -\sin \omega t \end{pmatrix} \right] \cos \frac{(2n+1)\pi}{2l} x$$

